TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

Topic: Customers Management System for NAWACO

Project Code: CMSN

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– Hanoi, Jan 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
02/12/2025	A	HoanTX	Add overview description about project
04/01/2026	A	HoanTX	Write project background, existing systems
04/01/2026	A	DungND	Add Business Opportunity, Software Product Vision, Project Scope & Limitations
05/01/2026	M	HoanTX	Modify the Software Product Vision, Project Scope & Limitations
05/01/2026	M	DangPH	Modify the Project Introduction Overview
08/01/2026	M	HoanTX	Modify Topic name
10/01/2026	M	DangPH	Modify the Limitations & Exclusions
10/01/2026	A	GiangPT	Add Major feature
11/01/2026	M	DangPH, GiangPT	Modify the Major feature
14/01/2026	M	HoanTX	Modify the Major feature
15/01/2026	M	DangPH	Modify the Major feature
17/1/2026	M	HoanTX	Modify document based on new template
17/01/2026	M	DangPH	Modify document about water meter customers
19/01/2026	M	HoanTX	Replace features name in the 5.1 index
28/01/2026	M	GiangPT	Modify 3.Existing Solutions

*A - Added M - Modified D - Deleted

II. Definition and Acronyms

Acronym	Definition
CMSN	Customers Management System for NAWACO
NAWACO	Nam Dinh Water Supply Joint Stock Company
CL	Company Leadership
PTH	Planning Technical Head
BH	Business Head
CH	Construction Head
SS	Survey Staff
ORS	Order Receiving Staff
FS	Finance Staff
MIS	Meter Inspection Staff
CS	Construction Staff
IS	IT Staff
BR	Business Rules
BA	Business Analysis
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
PM	Project Manager
SRS	Software Requirement Specification
SDS	Software Design Specification
UT	Unit Test
IT	Integration Test
ST	System Test
UAT	User Acceptance Test
UC	Use Case
API	Application Program Interface

III. Project Introduction

1. Overview

1.1 Project Information

- English name: ***Customers Management System for NAWACO***
- Vietnamese name: ***Hệ thống quản lý khách hàng cho NAWACO***
- Project code: ***CMSN***
- Group name: ***SEP490-G39***
- Software type: Web Application and Mobile Application

1.2 Project Purpose

The purpose of the Customers Management System for NAWACO (CMSN) project is to design and develop a modern, robust, and user-friendly digital platform to replace the company's legacy and dysfunctional EOS-CRM system. This new system aims to comprehensively manage the end-to-end lifecycle of customer water service requests—from initial installation application to ongoing meter reading and billing—while enforcing standardized business workflows across the Planning-Technical, Finance, IT, Construction, and Business departments.

Key objectives include:

- Eliminating Operational Inefficiencies: Resolving critical bugs, incorrect business logic, and performance issues of the old system to ensure accurate data processing, reliable report generation, and swift task completion.
- Streamlining Core Business Processes: Digitizing and automating the workflow for new water meter installations, maintenance, meter recording, and billing to reduce manual paperwork, minimize errors, and accelerate service delivery.
- Enhancing Data Integrity and Accessibility: Providing a single source of truth for customer information, project status, and infrastructure data, with secure role-based access control for all departments.
- Improving Interdepartmental Collaboration: Integrating real-time communication tools such as notifications functionalities within the platform to facilitate seamless information exchange and coordination between teams.
- Laying a Foundation for Future Digitalization: Building a scalable and maintainable system architecture that can support future enhancements such as electronic contracts, e-invoices, and advanced analytics for smart water management.

Ultimately, the CMSN seeks to empower NAWACO employees with a reliable tool that improves daily productivity, ensures regulatory compliance, and elevates the overall quality of service provided to the citizens of Nam Dinh.

1.3 Project Stakeholders

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Table 1: Project Stakeholders

2. Product Background

NAWACO is a company responsible for managing water projects and providing clean water resources for citizens of Nam Dinh City (now Nam Dinh Ward, Ninh Binh Province). Employees have been using an internal web application for their work; however, for many reasons, this application contains numerous bugs and errors that have not been fixed. In addition, there are no documents, no guidelines, and no source code available. As a result, the company cannot improve the application and is forced to continue using it despite the existing errors.

Our mission is to create a new platform that fixes existing errors, adds new features, and enhances performance. The new platform will support the business operations of several departments, such as the Planning-Technical Department, IT Department, Business Department and Construction Department.

In the new platform, we focus on a single business process: improving the customer experience in using clean water services, with the following steps:

- Customers come to the Planning-Technical Department to write an installation request form and submit it to an employee who has permission to receive the form
- The employee enters customer information into the platform.
- The Department Head assigns an employee to go to the installation location to conduct a site survey.
- The department creates a project cost estimation and requests approval from the Planning-Technical Department Head. After approval, the estimation is assigned by a company leader and sent to the Finance Department.
- The Finance Department checks the estimation, prints a bill, and requests payment from the customer.

- After the customer pays the estimation, the Planning-Technical Department hands the profile to the Construction Department to carry out the construction work, create a settlement, and send it to the Finance Department.
- The Finance Department requests the customer to pay the remaining balance if the settlement is higher than the estimation, or refunds the difference if it is lower.
- Create a new customer, save him/her and maintain construction; meter recording and billing per month.

3. Existing Solutions

Currently, employees of **NAWACO** are using a platform named **EOS-CRM**, which includes a **web-based system** and a **mobile application** supporting operational activities.

3.1. Short description about the platform:

- Name: EOS-CRM
- Link: http://113.160.196.170:88
- Technology: ASP.NET

3.2. Services and Strengths

For admin feature

- **Overall Device Management**: The system supports managing supplies, the company's organizational structure, and information about the clean water system in the former Nam Dinh Province (including water meters, laterals, administrative units,...). When decrees or regulations are issued by the State or the provincial People's Committee, staff from the Computer Department can easily update indexes, add new items, or remove obsolete data.

For general feature

- **Customer Management**: All customers using clean water services are stored in the system and supported when issues occur.
- **Installation Water Meter Construction Initialization**: The company can create new projects and execute business processes accurately, from receiving installation requests to managing completed projects.
- **Reports Exporting**: Employees can export or print reports related to estimation, settlement, and supplies when needed

For mobile feature

- **Water Meter Reading Recording**:
  The mobile application allows staff to record water meter readings at the field using mobile devices.

- **Meter Image Capturing**:
  Staff can capture and upload images of water meters to support verification and review of recorded readings.
- **Debt Information Lookup**:
  The application supports checking customer debt information to assist field operations.

Weakness

- **Performance**:
  When employees generate reports, the preview sometimes loads slowly. In particular, the supplies report modal often displays a loading spinner instead of previewing the report document despite having only 141 records.
- **Poor Design**:
  When reporting customer water meter indexes, the preview omits all records in the price column. Additionally, there are many unnecessary fields that are not required for business operations, such as "Loại ĐT" on the run-estimation page (employees do not understand why it exists). The website is also not responsive.
- **Inability to Update Data**:
  Due to the lack of support from the programming team, some indexes are outdated after new regulations are issued. As a result, employees must manually enter updated indexes.
- **Incorrect Business Logic Implementation**:
  In several billing periods, the platform calculates water prices twice. Consequently, the company must refund customers or notify them of billing errors.
- **Manual Operations in Mobile Application**:
  Although the mobile application supports meter reading activities, some tasks still require manual input, such as recording water meter readings. This increases workload and may lead to data inconsistency and human errors.

4. Solution & Opportunity

4.1. Solution

The Customers Management System for NAWACO (CMSN) is designed to provide a comprehensive and unified solution for managing clean water services across all communes and wards within the former Nam Dinh Province. The system aims to streamline and standardize the entire business lifecycle—from customer intake to service operation and billing—by emphasizing efficiency, transparency, and data integrity. CMSN supports core business operations, including water meter installation and maintenance, customer management, clean water service management, and debt tracking. These objectives are achieved by establishing end-to-end automated workflows that cover all operational stages, minimizing manual processing, reducing data entry errors, and improving the overall efficiency of construction management and service delivery.

To enhance interdepartmental collaboration, the system integrates real-time communication features such as instant messaging and centralized notifications. This enables staff across different departments and branches to work effectively and stay informed without switching between multiple applications. These communication capabilities can also be extended to support customer service operations.

In addition, CMSN modernizes water fee collection through a robust mobile application. Key features include water meter reading, automatic tariff calculation, abnormal consumption detection, water meter image capture, and automated payment notifications sent to customers.

In the future, the system will adopt electronic signatures, electronic invoices, and electronic contracts to ensure legal validity and full regulatory compliance. By standardizing document processing workflows and ensuring data integrity, CMSN prevents data forgery and unauthorized modifications. The platform also delivers a secure and user-friendly online contract signing experience, with a clear and transparent user interface, significantly enhancing accessibility and the overall user experience.

4.2. Opportunity

4.2.1 Current Challenges

NAWACO is currently facing significant challenges arising from its existing management system. The internal web application, EOS-CRM, has become obsolete and is burdened with numerous unresolved technical issues. The absence of system documentation, lack of access to source code, and the unavailability of technical support from the original development team have made it impossible for the company to maintain, enhance, or upgrade the system.

As a result, employees across departments are forced to continue using a highly defective system, leading to decreased operational efficiency, prolonged processing times, and a direct negative impact on the quality of customer service.

Specific issues include incorrect implementation of business logic, a non-intuitive and non-responsive user interface, slow system performance—particularly during report generation, inability to update meter readings in accordance with new regulations, and the lack of features to support modern, standardized workflows.

4.2.2 Market Opportunity

These challenges simultaneously present a significant opportunity for comprehensive digital transformation. We propose a complete replacement of the legacy system with a modern water supply project management platform, specifically designed to meet NAWACO's business requirements.

The proposed solution will not only resolve all existing system deficiencies but will also introduce advanced functionalities, optimize performance, and fully support cross-departmental business processes.

Market opportunities include the full digitization of the clean water installation process from request intake to project completion; standardization of business operations across the Plan–Technical, Finance, and Construction departments; establishment of a scalable foundation to expand services to additional regions in the future; and the application of modern technologies to enhance water resource management.

4.2.3 Strategic Value

The project delivers substantial strategic value to NAWACO by streamlining the processing of water installation requests to reduce turnaround time and paperwork; strengthening interdepartmental collaboration by eliminating information silos and enabling seamless workflows; providing real-time project tracking to ensure comprehensive visibility into project progress and status; automating workflow management to minimize manual intervention and improve accuracy; and establishing a robust foundation for smart water management, supporting future applications and data-driven analytics.

5. Project Scope & Limitations

The CMSN system will provide a comprehensive solution for managing and monitoring the lifecycle of clean water system construction and customer management. It will be used internally through automated workflows with real-time tracking capabilities covering all operational stages. In addition, built-in notification mechanisms are designed to support core business operations, eliminating the need for third-party applications. The software will include features such as water fee collection, payment information visualization, and electronic signatures in compliance with government digitalization regulations.

However, the system will not include features related to online payments, electronic contract signing, electronic invoicing, or debt tracking. Any requests or features beyond this defined scope will not be considered unless they deliver significant value and receive approval from NAWACO's Project Managers and company leadership, along with the necessary adjustments to budget, timeline, and resources.

5.1 Major Features

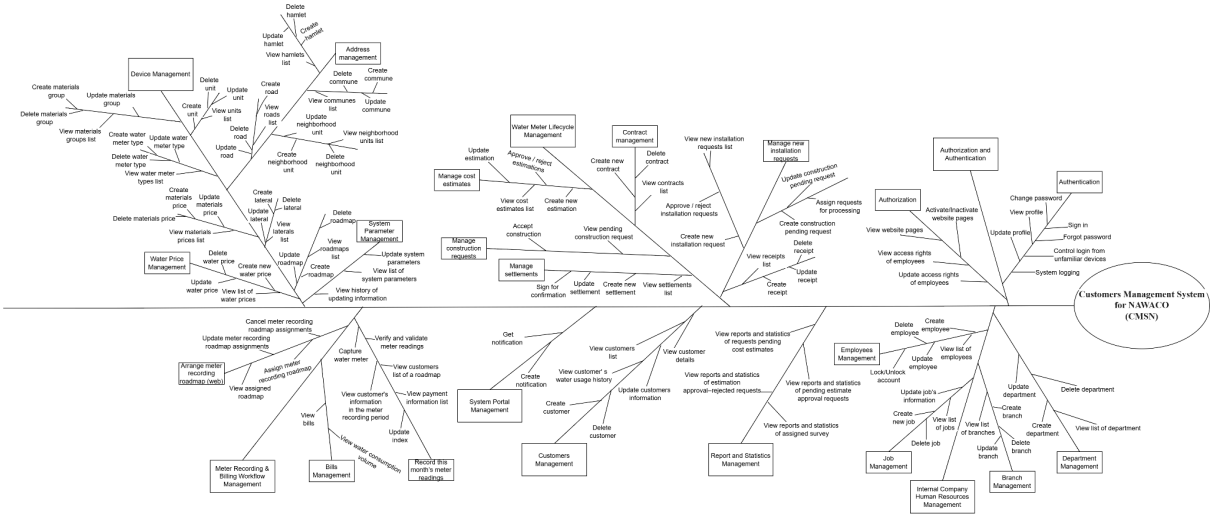

Figure 1: Feature Tree

[View image details]

Feature ID	Feature
Business Administrator/IT Department Features	
FE-01	**Authorization and Authentication:** Covers access control functions based on users and departments; Covers user authentication functions such as login and session management; Allow employees in the IT Department to login, reset password if forgotten or necessary
FE-02	**Report and Statistics Management:** Enables business administrators to view statistics about devices, water customers, internal human resource, system portal and customer's water meter lifecycle management.
FE-03	**Customers Management:** Enables business administrators to view, update, and delete customers that have been using water service.
FE-04	**System Portal Management:** Enables business administrators to view, create work notifications for overall employees in NAWACO company

FE-05	**Internal Human Resource Management:** Enables business administrators to view, update, create and delete employees' information, their departments, branches and jobs
FE-06	**Meter Recording and Billing Management:** Enables business administrators to view the meter indexes of customers in the current month or previous months; view, create, update and delete billing for water price payment
FE-07	**Customer's Water Meter Lifecycle Management:** Enables business administrators to facilitate creation, update and approval/rejection of new installation request (form), estimation, contract, new customers
FE-08	**Device Management**: Enables business administrators to view, create, update and delete devices that NAWACO company have such as overall water meter, water meter, supplies,...
Planning and Technical Department Features	
FE-01	**Customer's Water Meter Lifecycle Management:** Allow employees in the Planning and Technical Department to view, create, update, approve/reject new installation requests, cost estimates, water meter maintenance operations, contract; accept construction.
FE-02	**Customers Management:** Allow employees in the Planning and Technical Department to create, view, update and delete customers' information, include of their usage histories
FE-03	**System Portal Management:** Allow employees in the Planning and Technical Department to view notifications
FE-04	**Authentication and Authorization**: Allow employees in the Planning and Technical Department to login, reset password if forgotten or necessary
Finance Department Features	

FE-01	**Customer's Water Meter Lifecycle Management:** Allow employees in the Finance Department to view, sign for confirming cost estimations and settlements.
FE-02	**System Portal Management:** Allow employees in the Finance Department to view notifications
FE-03	**Authentication and Authorization**: Allow employees in the Finance Department to login, reset password if forgotten or necessary
Construction Department Features	
FE-01	**Customer's Water Meter Lifecycle Management:** Allow employees in the Construction Department to view, create, update, approve/reject new settlements; view, approve/reject construction pending request.
FE-02	**System Portal Management:** Allow employees in the Construction Department to view notifications
FE-03	**Authentication and Authorization**: Allow employees in the Construction Department to login, reset password if forgotten or necessary
Sale Department Features	
FE-01	**Meter Recording & Billing Workflow Management:** Allow employees in the Sale Department to record water meter index of each customers of their roadmaps and calculate water price bills
FE-02	**System Portal Management:** Allow employees in the Sale Department to view notifications
FE-03	**Authentication and Authorization**: Allow employees in the Sale Department to login/logout, reset password if forgotten or necessary

Table 2: Major Feature

5.2 Limitations & Exclusions

5.2.1 Limitations

ID	Name	Description
LI-01	Legacy System Integration	No integration with existing buggy system; complete replacement only
LI-02	Offline Capabilities	Minimal offline functionality; primarily online platform
LI-03	Customer Self-Service Portal	Customers cannot directly access system; employee-mediated only
LI-04	Multi-language Support	Vietnamese-only interface in initial release
LI-05	Social Media Integration	No social media connectivity or marketing features

Table 3: Limitations

5.2.2 Exclusions

ID	Name	Description
EX-01	External Contractor Management	Only NAWACO internal teams managed
EX-02	HR/Payroll Functions	The system does not include functionalities related to Human Resources (HR) and Payroll
EX-03	Payment Functions	No payment resolving

Table 4: Exclusions